Quizzes

Q1

D Which error detection method uses one's complement arithmetic?

A) Simple parity check

B) Two-dimensional parity check

C) CRC

D) Checksum

Q2

A Which error detection method consists of just one redundant bit per data unit?

A) Simple parity check

B) Two-dimensional parity check

C) CRC

D) Checksum

Q3

In cyclic redundancy checking, what is the CRC?

A) The divisor

B) The quotient

C) The dividend

D) The remainder

Q4

B A _____ error means that two or more bits in the data unit have changed.

A) double-bit

B) burst

C) single-bit

D) none of the above

Q5

C In _____ error correction, the receiver corrects errors without requesting retransmission.

向前纠错技术：

可使接收端纠错，而无需重传

A) backward

B) onward

C) forward

D) none of the above

Q6

B In _____ error correction, the receiver asks the sender to send the data again.

A) backward

B) retransmission

C) forward

D) none of the above

Q7

模二运算

C In modulo-2 arithmetic, _____ give the same results.

A) addition and multiplication

B) addition and division

C) addition and subtraction

D) none of the above

Q8

A In modulo-2 arithmetic, we use the _____ operation for both addition and subtraction.

A) XOR

B) OR

C) AND

D) none of the above

Q9

分组码

C In _____ coding, we divide our message into blocks, each of k bits, called ___.

A) block; blockwords

B) linear; datawords

C) block; datawords

D) none of the above

Q10

C We add r redundant bits to each block to make the length n = k + r. The resulting n-bit blocks are called _____.

A) datawords

B) blockwords

C) codewords

D) none of the above

Q11

B The _____ between two words is the number of differences between corresponding bits.

A) Hamming code

B) Hamming distance

C) Hamming rule

D) none of the above

Q12

B To guarantee the detection of up to 5 errors in all cases, the minimum Hamming distance in a block code must be _____.

A) 5

B) 6

C) 11

D) none of the above

检测的最大错误数 = dmin − 1

Q13

A

In a linear block code, the _____ of any two valid codewords creates another valid codeword.

A) XORing

B) ORing

C) ANDing

D) none of the above

Q14

D

A simple parity-check code can detect _____ errors.

简单奇偶检测只能检测奇数个错误

A) an even-number of

B) two

C) no errors

D) an odd-number of

Q15

C

_____codes are special linear block codes with one extra property. If a codeword is rotated, the result is another codeword.

A) Non-linear

B) Convolution

C) Cyclic 看环冗余编码：将码字旋转后得到的还是码字

D) none of the above

Q16

A The _____of errors is more difficult than the _____.

A) correction; detection

B) detection; correction

C) creation; correction

D) creation; detection

Q17

C In modulo-2 arithmetic, we use only _____.

A) 1 and 2

B) 0 and 2

C) 0 and 1

D) none of the above

Q18

C Adding 1 and 1 in modulo-2 arithmetic results in _____.

A) 1

B) 2

C) 0

D) none of the above

Q19

C The Hamming distance between equal codewords is _____.

A) 1

B) n

C) 0

D) none of the above

Q20

A The Hamming distance between 100 and 001 is _____.

A) 2

B) 0

C) 1

D) none of the above

Q21

B Checksums use _____ arithmetic.

A) two's complement arithmetic

B) one's complement arithmetic

C) either (a) or (b)

D) none of the above

Q22

B The checksum of 1111 and 1111 is _____.

A) 1111

B) 0000

C) 1110

D) 0111

① 1111 + 1111 = 11110

② 回卷机位 1111 0

③ 1111 再取反

④ 0000

Q23

A The checksum of 0000 and 0000 is _____.

A) 1111

B) 0000

C) 1110

D) 0111

① 0000 + 0000 = 0000

② 回卷机位 0000 再取反

③ 1111

Revision Questions

Q1.

How does a single-bit error differ from a burst error?

In a single-bit error only one bit of a data unit is corrupted; in a burst error more than one bit is corrupted (not necessarily contiguous).

Q2.

Discuss the concept of redundancy in error detection and correction.

Redundancy is a technique of adding extra bits to each data unit to determine the accuracy of transmission.

Q3

Distinguish between forward error correction versus error correction by retransmission.

In forward error correction, the receiver tries to correct the corrupted codeword; in error detection by retransmission, the corrupted message is discarded (the sender needs to retransmit the message).

Q4

What is the definition of a linear block code? What is the definition of a cyclic code?

A linear block code is a block code in which the exclusive-OR of any two codewords results in another codeword.

A cyclic code is a block code in which the rotation of any codeword results in another codeword.

Q5

What is the Hamming distance? What is the minimum Hamming distance?

The Hamming distance between two words (of the same size) is the number of differences between the corresponding bits. The Hamming distance can easily be found if we apply the XOR operation on the two words and count the number of 1s in the result. The minimum Hamming distance is the smallest Hamming distance between all possible pairs in a set of words.

Q6

How is the simple parity check related to the two-dimensional parity check?

The single parity check uses one redundant bit for the whole data unit. In a two dimensional parity check, original data bits are organized in a table of rows and columns. The parity bit is then calculated for each column and each row.

Q7

What kind of arithmetic is used to add data items in checksum calculation?

One's complement arithmetic is used to add data items in checksum calculation. In this arithmetic, when a number needs more than n bits, the extra bits are wrapped and added to the number. In this arithmetic, the complement of a number is made by inverting all bits.

Q8

- What kind of error is undetectable by the checksum?
- At least three types of error cannot be detected by the current checksum calculation. First, if two data items are swapped during transmission, the sum and the checksum values will not change.
- Second, if the value of one data item is increased (intentionally or maliciously) and the value of another one is decreased (intentionally or maliciously) the same amount, the sum and the checksum cannot detect these changes.
- Third, if one or more data items is changed in such a way that the change is a multiple of 216 − 1, the sum or the checksum cannot detect the changes.

Q9

List the three main operations that can be performed on a message (e.g., a file, an image, a document, etc). How do they affect the size of the message?

- A message can be encoded (its size increases) and be made more robust to channel noise.
- It can be compressed (its size drops) to reduce its space requirements.
- It can also be encrypted to make its content more secure from eavesdroppers, in which case its size can either be reduced, increased or remain the same, depending on the applied encryption rules.

Q10

How do channel coding and modulation affect data reliability and data rate (throughput)?

- Channel coding increases data reliability but reduces data rate.
- High-order modulation can increase data rate at the expense of data reliability.
- Combining modulation and coding schemes allows us to adjust their trade-offs depending on the application.

Q1

.What is the maximum effect of a 2-ms burst of noise on data transmitted at the following rates?

a. 1500 bps

b. 12 kbps

c. 100 kbps

d. 100 Mbps

a. vulnerable bits = $(1{,}500) \times (2 \times 10{-}3) = 3$ bits

b. vulnerable bits = $(12 \times 103) \times (2 \times 10{-}3) = 24$ bits

c. vulnerable bits = $(100 \times 103) \times (2 \times 10{-}3) = 200$ bits

d. vulnerable bits = $(100 \times 106) \times (2 \times 10{-}3) = 200{,}000$ bits

Q2

Q2. Exclusive-OR (XOR) is one of the most used operations in the calculation of codewords. Apply the exclusive-OR operation on the following pairs of patterns. Interpret the results.

a. $(10001) \oplus (10001)$

b. $(11100) \oplus (00000)$

c. $(10011) \oplus (11111)$

a. $(10001) \oplus (10001) = 00000$

b. $(11100) \oplus (00000) = 11100$

c. $(10011) \oplus (11111) = 01100$

Q3

Using the code in Table below, what is the dataword if each of the following

codewords is received?

a. 01011

b. 11111

c. 00000

d. 11011

Dataword	*Codeword*	*Dataword*	*Codeword*
0000	**00000**	1000	**10001**
0001	**00011**	1001	**10010**
0010	**00101**	1010	**10100**
0011	**00110**	1011	**10111**
0100	**01001**	1100	**11000**
0101	**01010**	1101	**11011**
0110	**01100**	1110	**11101**
0111	**01111**	1111	**11110**

a. error b. error c. 0000 d. 1101

Q4

What is the Hamming distance for each of the following codewords?

a. d (10000, 00000)

b. d (10101, 10000)

c. d (00000, 11111)

d. d (00000, 00000)

a. d (10000, 00000) = 1 b. d (10101, 10000) = 2

c. d (00000, 11111) = 5 d. d (00000, 00000) = 0

Q5

Assuming even parity, find the parity bit for each of the following data units.

a. 1001011
b. 0001100
c. 1000000
d. 1110111

Q6

We need to add all bits modulo-2 (XORing). However, it is simpler to count the number of 1s and make them even by adding a 0 or a 1. We have shown the parity bit in the codeword in colour and separated for emphasis.

Dataword	Number of 1s	Parity	Codeword
a. 1001011 ->	4 (even) ->	0	10010110
b. 0001100 ->	2 (even) ->	0	00011000
c. 1000000 ->	1 (odd) ->	1	10000001
d. 1110111 ->	6 (even) ->	0	11101110

Q7

Given the dataword 101001111 and the divisor 10111, show the generation of the CRC codeword at the sender site (using binary division).

Answer

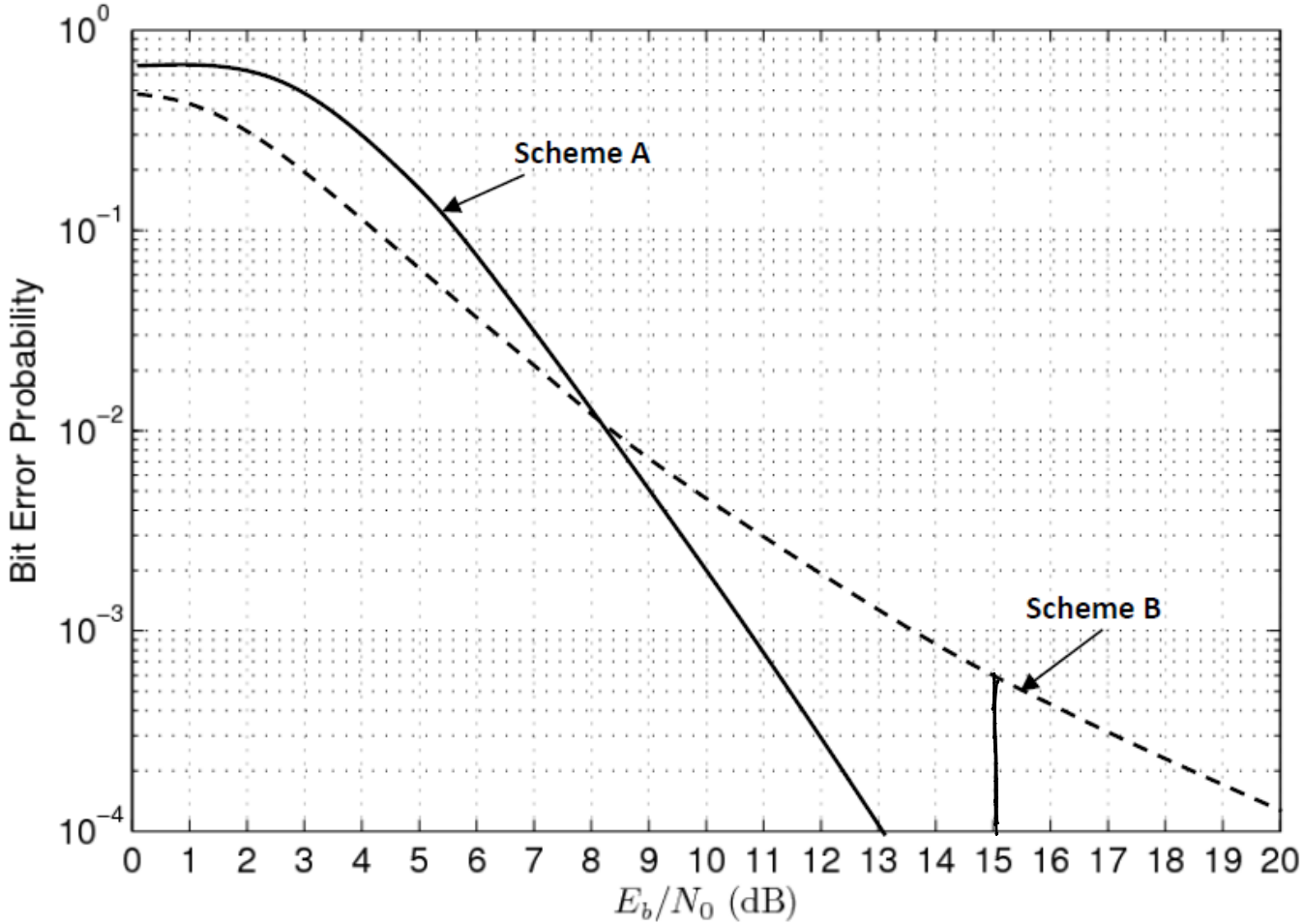
```
Dataword  1 0 1 0 0 1 1 1 1

                  1 0 0 1 1 0 1 1 1 Quotient
Divisor 1 0 1 1 1 | 1 0 1 0 0 1 1 1 1 0 0 0 0
                  1 0 1 1 1
                    0 0 1 1 1
                    0 0 0 0 0
                      0 1 1 1 1
                      0 0 0 0 0
                        1 1 1 1 1
                        1 0 1 1 1
                          1 0 0 0 1
                          1 0 1 1 1
                            0 1 1 0 0
                            0 0 0 0 0
                              1 1 0 0 0
                              1 0 1 1 1
                                1 1 1 1 0
                                1 0 1 1 1
                                  1 0 0 1 0
                                  1 0 1 1 1
                        Remainder 0 1 0 1

Codeword  1 0 1 0 0 1 1 1 1 0 1 0 1
```

The performance of two transmission schemes (A and B) is depicted in the figure below. Answer the following questions:

1. Which scheme is **better**?
2. If we transmit 5000 bits using Scheme B at Eb/N0 = 15 dB, how many **bits** should we expect to receive in **error**?
3. In order to achieve a bit error probability of 5x10-3 using Scheme A, how **many times** should the energy per bit Eb be higher than the power spectral density N0?
4. Assume that Scheme A corresponds to an **uncoded modulation** scheme that uses **8 symbols**. What is the required **SNR** (in dB) to achieve a bit error probability of 10-4?



Answer

a. Which scheme is **better**?

- Scheme B is better for low Eb/N0 values (up to 8 dB), whilst Scheme A is better for Eb/N0 values greater than 8 dB.

b. If we transmit 5000 bits using Scheme B at Eb/N0 = 15 dB, how many **bits** should we expect to receive in **error**?

- From the plot, we see that the bit error probability of Scheme B at Eb/N0 = 15 dB is 6x10-4. Therefore, when we transmit 5000 bits, we expect 5000x6x10-4 = 3 bits to be received in error.

Cont'd

a. In order to achieve a bit error probability of 5x10-3 using Scheme A, how **many times** should the energy per bit Eb be higher than the power spectral density N0?

- A bit error probability of 5x10-3 can be achieved for Eb/N0 = 9 dB when Scheme A is used. If we convert the value of 9 dB to a unitless number, i.e., 10(9dB/10) = 7.94, we find that Eb should be about 8 times higher than N0.

b. Assume that Scheme A corresponds to an **uncoded modulation** scheme that uses **8 symbols**. What is the required **SNR** (in dB) to achieve a bit error probability of 10-4?

- When Scheme A is employed, a bit error probability of 10-4 can be achieved for Eb/N0 = 13 dB. If Scheme A has 8=23 symbols available, we understand that 3 bits form a symbol. Therefore, the required SNR is 13 dB + 10log10(3) = 17.77 dB.

Thank you

Thank you